Visual Image Interpretation

Aerial photographs as well as imagery, obtained by remote sensing using aircraft or spacecraft as platforms, have applicability in various fields.

By studying the qualitative as well as quantitative aspects of images recorded by various sensor systems,

like aerial photographs (black-and-white, black-and-white infrared, colour and colour infrared),

multiband photographs, satellite data (both pictorial and digital) including thermal and radar imagery,

an interpreter well experienced in his field can derive a lot of information.

Image interpretation is defined as the act of examining images to identify objects and judge their significance.

An interpreter studies remotely sensed data and attempts through logical process to detect, identify, measure and evaluate the significance of environmental and cultural objects, patterns and spatial relationships.

It is an information extraction process.

Aerial photographs, as well as imagery, obtained by remote sensing employing electromagnetic energy as the means of detecting and measuring target/objects characteristics, has applicability to various fields because of four basic reasons.

First - It represents a larger area of the earth from a perspective view and provides a format that facilitates the study of objects and their relationships.

Second - Certain types of imagery and aerial photograph can provide a 3-D view.

Third - Characteristics of objects not visible to the human eye can be transformed into images

Fourth - It provides the observer with a permanent record/representation of objects at any moment of time. In addition, data is real-time, repetitive and, when in digital form, is computer compatible for quick analysis.

Factors Governing the Quality of an image

In addition to the inherent characteristics of an object itself, the following factors influence image quality:

- Sensor characteristics (film types, digital systems)
- Season of the year and time of day
- Atmospheric effects
- Resolution of the imaging system and scale
- Image motion
- Stereoscopic parallax

Factors Governing Interpretability

1. Visual and mental acuity of the interpreter

2. Equipment and technique of interpretation

3. Interpretation keys, guides, manuals and other aids.

Recognizing targets is the key to interpretation and information extraction.

Observing the differences between targets and their backgrounds involves comparing different targets based on any, or all, of the visual elements of

- **tone,**
- **shape,**
- **size,**
- **pattern,**
- **texture,**
- **shadow, and**
- **association**.



Tone refers to the relative brightness or colour of objects in an image. Generally, tone is the fundamental element for distinguishing between different targets or features. Variations in tone also allows the elements of shape, texture, and pattern of objects to be distinguished.



Shape refers to the general form, structure, or outline of individual objects.

Shape can be a very distinctive clue for interpretation.

Straight edge shapes typically represent urban or agricultural (field) targets, while natural features, such as forest edges, are generally more irregular in shape, except where man has created a road or clear cuts.

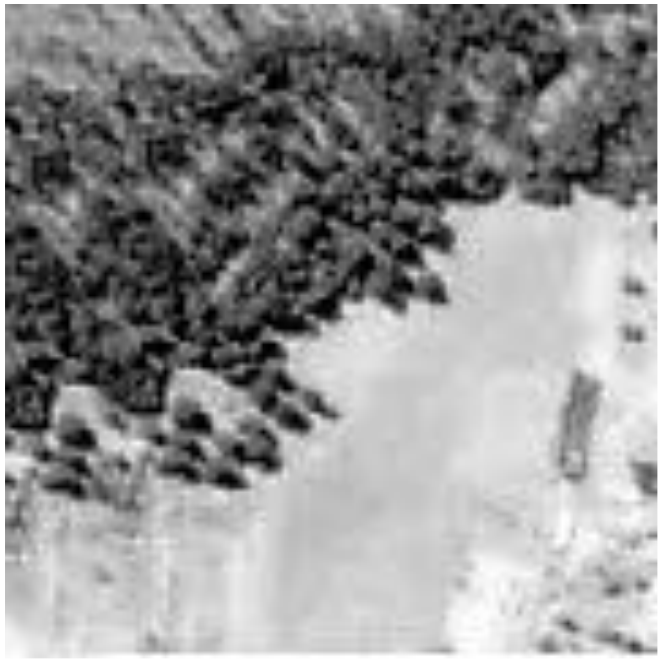
Farm or crop land irrigated by rotating sprinkler systems would appear as circular shapes.



Pattern refers to the spatial arrangement of visibly discernible objects.

Typically an orderly repetition of similar tones and textures will produce a distinctive and ultimately recognizable pattern.

Orchards with evenly spaced trees, and urban streets with regularly spaced houses are good examples of pattern.



Texture refers to the arrangement and frequency of tonal variation in particular areas of an image.

Rough textures would consist of a mottled tone where the grey levels change abruptly in a small area, whereas smooth textures would have very little tonal variation.

Smooth textures are most often the result of uniform, even surfaces, such as fields, asphalt, or grasslands.

A target with a rough surface and irregular structure, such as a forest canopy, results in a rough textured appearance.

Texture is one of the most important elements for distinguishing features in radar imagery.



Shadow is also helpful in interpretation as it may provide an idea of the profile and relative height of a target or targets which may make identification easier.

However, shadows can also reduce or eliminate interpretation in their area of influence, since targets within shadows are much less (or not at all) discernible from their surroundings.

Shadow is also useful for enhancing or identifying topography and landforms, particularly in radar imagery.



Association takes into account the relationship between other recognizable objects or features in proximity to the target of interest.

The identification of features that one would expect to associate with other features may provide information to facilitate identification.

In the example given above, commercial properties may be associated with proximity to major transportation routes, whereas residential areas would be associated with schools, playgrounds, and sports fields.